# A counterexample to Stanley's conjecture on Smith forms of Jucys–Murphy elements

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**Abstract**

For a partition $\lambda$ of $n$, let $S^\lambda_\mathbb{Z}$ be the integral Specht module and $X_k = (1,k)+\cdots+(k-1,k)$ the $k$-th Jucys–Murphy element. Stanley conjectured in 2016 a specific formula for the invariant factors of $X_k$ on $S^\lambda_\mathbb{Z}$ in terms of the contents of the cells that $k$ occupies in standard Young tableaux of shape $\lambda$, and remarked, more weakly, that each is then a product of eigenvalues. Both statements are false. For $\lambda = (3,1,1)$ and $k = 3$ the Smith form is $(1,1,1,1,2,2)$, not the predicted $(1,1,1,1,1,4)$. For $\lambda = (5,2)$ and $k = 7$ the eigenvalues are 0 and 4, yet the Smith form contains a 2. Lean checks both matrix computations, and two independent programs check the matrices from the definitions. From the identity $X_3\big(X_3 - (1,2)\big) = 2$ in $\mathbb{Z}\mathfrak{S}_n$, checked in Lean, we prove in writing that for $n \geq 3$ the conjecture holds at $k = 3$ exactly when $\lambda_1 \leq 2$ or $\lambda$ has at most two rows, on $S^\lambda_\mathbb{Z}$ and on every full-rank lattice in $S^\lambda_\mathbb{Q}$ stable under the permutations of $\{1,2,3\}$.

The proofs, the Lean formalization, the certificate checkers and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

Status: Written proof, with Lean and computer checks; not yet reviewed by experts.

## 1 Introduction

In his survey *Smith normal form in combinatorics* [8], Richard P. Stanley proposed studying the Smith normal form (SNF) of natural elements of the group ring $\mathbb{Z}\mathfrak{S}_n$. His example is the Jucys–Murphy element $X_k = (1,k) + (2,k) + \cdots + (k-1,k)$ [4, 6], acting on the Specht module $S^\lambda$, where its eigenvalues are the contents of the cells occupied by $k$ in the standard Young tableaux of shape $\lambda$. Stanley reports computations through $n \leq 7$, which he credits to Darij Grinberg, and on that basis states Conjecture 7.4: if $W_{\lambda,k}$ has Smith form $(\alpha_1, \ldots, \alpha_{f^\lambda})$, then

$$\alpha_{f^\lambda - r + 1} = \pm \prod_{(i,j) \in S_r} (j - i) \qquad (1 \leq r \leq f^\lambda),$$

where $f^\lambda$ is the number of standard Young tableaux of shape $\lambda$ and $S_r$ is the set of cells $(i,j)$ that $k$ occupies in at least $r$ of them. He adds that every SNF entry is then a product of some of the eigenvalues. Appendix A quotes the passage in full; Stanley's example is $\lambda = (5,1)$, $k = 5$, where the formula gives $(1,1,3,3,12)$.

On December 25, 2024, Alexander Chervov asked on MathOverflow about the status of these conjectures [1]; the question had no answer as of October 21, 2026. In a comment on it, Grinberg noted that the survey defines the matrices $W_{\lambda,k}$ "somewhat clumsily", through a basis of $\mathbb{Z}\mathfrak{S}_n$, and

clarified the intended lattice: the integral Specht module with its standard polytabloid basis. We use exactly this reading (Section 2).

## Results

**Theorem 1** (Smith form checked in Lean; Specht-module matrix checked by two certificate checkers)**.** *For $\lambda = (3,1,1)$ and $k = 3$, the Smith normal form of $X_3$ acting on the integral Specht module $S^{(3,1,1)}_{\mathbb{Z}}$ is $(1,1,1,1,2,2)$. Conjecture 7.4 predicts $(1,1,1,1,1,4)$. Hence Conjecture 7.4 is false.*

Supporting checks: Lean checks the Smith form of the explicit matrix $A$ of Section 3 and that the predicted form is unreachable from it. The identification of $A$ is integral: checker A proves that the standard polytabloids form a $\mathbb{Z}$-basis of the span of all polytabloids and that $A$ is the integer matrix of $X_3$ in that basis, and checker B verifies the exact equations $X_3 e_{T_j} = \sum_i A_{ij} e_{T_i}$ in the tabloid module. The two checkers are independent: separate code, different enumerations of the standard tableaux, and different arithmetic (Appendix D).

**Proposition 2** (Smith form and eigenvalues checked in Lean; Specht-module matrix checked by two certificate checkers)**.** *Let $\lambda = (5,2)$ and $k = 7$, so $n = 7$ and $f^\lambda = 14$. On $S^{(5,2)}_{\mathbb{Q}}$, $X_7(X_7 - 4) = 0$, and the eigenvalues of $X_7$ are $0$, with multiplicity $5$, and $4$, with multiplicity $9$. The Smith normal form of $X_7$ on $S^{(5,2)}_{\mathbb{Z}}$ is*

$$(1,1,1,1,2,4,4,4,4,0,0,0,0,0),$$

*while Conjecture 7.4 predicts $(1,1,1,1,1,4,4,4,4,0,0,0,0,0)$. The invariant factor $2$ is not a product of eigenvalues, so Stanley's weaker remark also fails, for a two-row shape with $k = n$.*

Supporting checks: for an explicit $14 \times 14$ integer matrix $A$ (given in `data/witness-52-k7.json`), Lean proves its Smith form via unimodular matrices (`smith_form_52`, `U52_unimodular`, `V52_unimodular`), the relation $A(A - 4I) = 0$ (`eigen_identity_52`), its characteristic polynomial $x^5(x-4)^9$, which gives the multiplicities 5 and 9 (`charpoly_52`), that its rational eigenvalues are exactly 0 and 4 (`eigenvalues_52`), and that every unimodular diagonalization of it has an entry divisible by 2 but not by 4 (`weak_remark_fails_52`); `counterexample_52` combines the Smith form, the eigenvalues and the last statement. Two independent certificate checkers (Appendix D) verify, from the definitions, that this same $A$, with the same tableau order and column convention, is the matrix of $X_7$ in the standard polytabloid basis of $S^{(5,2)}_{\mathbb{Z}}$. From $\lambda$ and $k$ they take as eigenvalues the contents of the cell holding 7 in each standard tableau, verify $\prod_c (A - cI) = 0$ over the distinct contents $c$, check the trace identities $\operatorname{tr}(A^j) = \sum c^j$ (over the eigenvalue list) for $j = 1, \dots, r-1$, where $r$ is the number of distinct contents (for $j = 0$ it is immediate), which fix the multiplicities, and find a Smith entry that is not the absolute value of a product of a sub-multiset of the eigenvalues.

Both counterexamples sit inside the range $n \le 7$ of the computations that Stanley reports. At $k = 3$ the failures follow a pattern, which rests on an identity in $\mathbb{Z}\mathfrak{S}_3 \subseteq \mathbb{Z}\mathfrak{S}_n$, checked in Lean (Lemma 5):

$$X_3(X_3 - (1,2)) = (X_3 - (1,2))X_3 = 2.$$

It forces every invariant factor of $X_3$ to divide 2, on every lattice stable under the permutations of $\{1,2,3\}$. The next two results are proved in writing and are not machine-checked.

**Theorem 3** (proved in writing; not machine-checked)**.** *Let $\lambda \vdash n \ge 3$, and let $m$ be the number of standard Young tableaux of shape $\lambda$ in which $3$ occupies the cell $(1,3)$ or $(3,1)$. On every lattice in $S^\lambda_{\mathbb{Q}}$ that is stable under the permutations of $\{1,2,3\}$, in particular on $S^\lambda_{\mathbb{Z}}$, the Smith normal form of $X_3$ is $(1^{f^\lambda - m}, 2^m)$.*

Supporting checks: Lean proves the identity of Lemma 5, and that every unimodular diagonalization of an integer matrix $A$ with $AB = 2I$ for an integer matrix $B$ has diagonal entries in $\{\pm 1, \pm 2\}$, with $v_2(\det A)$ of them even. Applying this to the nonnegative Smith form is a written step. The census confirms the conclusion on $S^\lambda_\mathbb{Z}$ in all 63 cases with $k = 3$ and $n \le 8$.

**Corollary 4** (proved in writing; not machine-checked)**.** *Let $\lambda \vdash n \ge 3$. Conjecture 7.4 holds at $k = 3$ if and only if $\lambda_1 \le 2$ or $\lambda$ has at most two rows. When $\lambda_1 \ge 3$ and $\lambda$ has at least three rows, it fails on every lattice as in Theorem 3: the predicted largest invariant factor is 4, while every actual one divides 2.*

Supporting checks: Lean proves the final counting step of the proof; the census confirms the classification in all 63 cases with $k = 3$ and $n \le 8$, of which 23 fail.

No case with $n \le 4$ fails, and at $n = 5$ the only failures are $\lambda = (3, 1, 1)$ with $k = 3$ and with $k = 5$ (Proposition 7, an exact computation); in this sense these are the smallest counterexamples. A census of all $\lambda \vdash n \le 8$ and all $k$ finds 94 disagreements among 416 cases (Section 7), and the modifications of the formula we tested all fail (Appendix C).

**What is checked, and how.** The verified claims are Theorem 1 and Proposition 2: Lean checks the matrix computations, and two certificate checkers check, from the definitions of tableaux, tabloids and polytabloids, that the matrices are those of $X_k$ on the integral Specht lattices and that the predicted lists are Stanley's. Proposition 7, minimality on the integral Specht lattices for $n \le 5$, is a bounded exact computation by a third checker. Lean also proves Lemma 5, the matrix step of Theorem 3 and the counting step of Corollary 4; the remaining steps of those two proofs are written. Results labeled "proved in writing" are not machine-checked. The census for $n \le 8$ is supporting data, not part of any proof. Section 9 summarizes the checks and Appendix D gives their scope.

**Credits.** The conjecture, its formulation and its worked example are Stanley's. The computations on which it rests are reported in the survey as Grinberg's, and the clarification of the intended lattice is his. Chervov's question drew attention to the status of the conjecture. We know of no earlier report that the conjecture fails. We do not know why the computations reported in the survey supported the conjecture, and we attribute no error to them (Section 8).

## 2 The reading used

We follow Grinberg's clarification. Let $M^\lambda_\mathbb{Z}$ be the free $\mathbb{Z}$-module on the row tabloids $\{T\}$ of shape $\lambda$, with $\mathfrak{S}_n$ acting by $\pi\{T\} = \{\pi T\}$, where $\pi T$ replaces each entry $x$ of $T$ by $\pi(x)$. For a tableau $T$ with column stabilizer $C_T$, the polytabloid is $e_T = \sum_{\pi \in C_T} \operatorname{sgn}(\pi)\, \pi\{T\}$, and the integral Specht module $S^\lambda_\mathbb{Z}$ is the $\mathbb{Z}$-span of all $e_T$. The standard polytabloids $e_T$, $T$ standard, form a $\mathbb{Z}$-basis of $S^\lambda_\mathbb{Z}$ ([3, §8], [7, §2.6]); for the cases we compute, this is also verified directly by the checkers. Then $W_{\lambda,k}$ is the matrix of $X_k$ in the standard polytabloid basis, and its Smith form is that of $X_k$ as an endomorphism of $S^\lambda_\mathbb{Z}$, which does not depend on the basis.

By a lattice in $S^\lambda_\mathbb{Q}$ we mean a full-rank $\mathbb{Z}$-lattice, that is, a finitely generated $\mathbb{Z}$-submodule spanning $S^\lambda_\mathbb{Q}$ over $\mathbb{Q}$, equivalently a free $\mathbb{Z}$-module of rank $f^\lambda$ spanning $S^\lambda_\mathbb{Q}$; the Smith form of an endomorphism of a lattice is that of its matrix in any $\mathbb{Z}$-basis. The cell $(i, j)$ is in row $i$ and column $j$, and $j - i$ is its content. Smith forms are written with nonnegative entries in divisibility order, $\alpha_1 \mid \alpha_2 \mid \cdots$, so that any zeros come last; an empty product is 1. Where an unordered collection of invariant factors is meant, it is called a multiset.

## 3 The counterexample $\lambda = (3,1,1)$, $k = 3$

Let $\lambda = (3,1,1)$, so $n = 5$, $f^\lambda = 6$ and $X_3 = (1,3) + (2,3)$. Order the standard tableaux as

$$T_1 = \begin{array}{ccc}1&2&3\\4\\5\end{array}, \quad T_2 = \begin{array}{ccc}1&2&4\\3\\5\end{array}, \quad T_3 = \begin{array}{ccc}1&2&5\\3\\4\end{array}, \quad T_4 = \begin{array}{ccc}1&3&4\\2\\5\end{array}, \quad T_5 = \begin{array}{ccc}1&3&5\\2\\4\end{array}, \quad T_6 = \begin{array}{ccc}1&4&5\\2\\3\end{array},$$

and let column $j$ of $A$ hold the coordinates of $X_3 e_{T_j}$ in the standard polytabloid basis, so that $X_3 e_{T_j} = \sum_i A_{ij} e_{T_i}$. Let $B = 2A^{-1}$; by Lemma 5 it is the matrix of $X_3 - (1,2)$ in the same basis, and it is integral. Then

$$A = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & -2 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 2 & 0 & 0 \\ -1 & 0 & 2 & 0 & 2 & 0 \\ 0 & -1 & -1 & -1 & -1 & -1 \end{pmatrix},$$

and $AB = BA = 2I$. For example, the first column of $A$ says $X_3 e_{T_1} = 2e_{T_1} - e_{T_2} + e_{T_3}$.

**The prediction.** The entry 3 occupies the cell $(1,3)$ in $T_1$, $(2,1)$ in $T_2$ and $T_3$, $(1,2)$ in $T_4$ and $T_5$, and $(3,1)$ in $T_6$. Hence $S_1 = \{(1,2),(1,3),(2,1),(3,1)\}$ with $|\prod(j-i)| = |1\cdot 2\cdot(-1)\cdot(-2)| = 4$, $S_2 = \{(1,2),(2,1)\}$ with product of absolute value 1, and $S_r = \emptyset$ for $3 \le r \le 6$. Conjecture 7.4 therefore predicts $(1,1,1,1,1,4)$.

**The Smith form.** With

$$U = \begin{pmatrix} 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ -1 & 0 & 2 & 0 & 2 & 0 \\ 0 & -1 & -1 & -1 & -1 & -1 \end{pmatrix}, \qquad V = \begin{pmatrix} 1 & -1 & 1 & 1 & -1 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

one has $\det U = -1$, $\det V = 1$ and $UAV = \mathrm{diag}(1,1,1,1,2,2)$, which proves Theorem 1. Without transformation matrices: $A$ is lower triangular, so $\det A = 2\cdot(-1)\cdot(-1)\cdot 1\cdot 1\cdot(-2) = -4$. Modulo 2, the first row of $A$ vanishes and the sum of rows 2 to 6 vanishes, while rows 2 to 5 restricted to columns 2 to 5 form a unitriangular matrix; hence $A$ has rank 4 over $\mathbb{F}_2$. So exactly two invariant factors are even, their product with the others is 4, and the form is $(1,1,1,1,2,2)$. The predicted form has only one even factor and would require rank 5 over $\mathbb{F}_2$. The relation $AB = 2I$ gives the same conclusion a third way: every invariant factor divides 2, and $v_2(\det A) = 2$ of them equal 2. All three computations are checked in Lean (Section 9). This is also the case $m = 2$ ($T_1$ and $T_6$) of Theorem 3.

By Theorem 3 (proved in writing), the same Smith form arises on every lattice stable under the permutations of $\{1,2,3\}$ in a module isomorphic to $S^{(3,1,1)}_{\mathbb{Q}}$ (Appendix C). Appendix B derives $A$ by hand, without coordinates in the tabloid module.

## 4 The weak remark: $\lambda = (5,2)$, $k = 7$

Let $\lambda = (5,2)$ and $k = n = 7$, so $f^\lambda = 14$. In a standard tableau of shape $(5,2)$ the entry 7 is the largest, so it occupies a corner: $(1,5)$, of content 4, in the 9 tableaux obtained from the standard tableaux of shape $(4,2)$, or $(2,2)$, of content 0, in the 5 obtained from those of shape $(5,1)$. So the eigenvalues of $X_7$ on $S^{(5,2)}_{\mathbb{Q}}$ are 4, with multiplicity 9, and 0, with multiplicity 5, and since $X_7$ acts diagonally on Young's seminormal basis [4, 6], $X_7(X_7 - 4) = 0$ there. Proposition 2 does not rest on this classical fact: the relation $A(A - 4I) = 0$ for the explicit matrix $A$, and the eigenvalues, are checked in Lean and by both certificate checkers.

**The prediction.** With $N_{(1,5)} = 9$ and $N_{(2,2)} = 5$, the sets of Conjecture 7.4 are $S_r = \{(1,5),(2,2)\}$ for $r \le 5$, with product 0; $S_r = \{(1,5)\}$ for $6 \le r \le 9$, with product 4; and $S_r = \emptyset$ for $r \ge 10$. Hence the prediction is $\alpha_{14} = \cdots = \alpha_{10} = 0$, $\alpha_9 = \cdots = \alpha_6 = 4$ and $\alpha_5 = \cdots = \alpha_1 = 1$, that is, $(1,1,1,1,1,4,4,4,4,0,0,0,0,0)$.

**The Smith form.** The matrix of $X_7$ in the standard polytabloid basis, and unimodular $U$ and $V$ bringing it to $\mathrm{diag}(1,1,1,1,2,4,4,4,4,0,0,0,0,0)$, are given in the certificate `data/witness-52-k7.json`. The rank is 9, as the eigenvalues require, and $\alpha_5 = 2$. Every product of eigenvalues is 1, a power of 4, or 0, so 2 is not one of them: the weaker remark fails, and so does the conjecture itself. Since every unimodular diagonalization of the matrix has an entry divisible by 2 but not by 4, the failure does not depend on how the diagonal form is normalized. The conjugate shape $(2,2,1,1,1)$ with $k = 7$ behaves in the same way, and these two are the first failures of the remark in the census.

## 5 The Jucys–Murphy element $X_3$

**Lemma 5** (checked in Lean)**.** *In $\mathbb{Z}\mathfrak{S}_n$, $n \ge 3$, let $u = (1,2)$. Then $X_3(X_3 - u) = (X_3 - u)X_3 = 2$.*

*Proof.* Write $s = (1,3)$ and $t = (2,3)$, so $X_3 = s + t$. Let $c$ and $c^2$ be the two 3-cycles on $\{1,2,3\}$. Then $X_3^2 = s^2 + t^2 + st + ts = 2 + c + c^2$, since $st$ and $ts = (st)^{-1}$ are the two 3-cycles. Each of $su$, $tu$, $us$, $ut$ is a product of two distinct transpositions in $\mathfrak{S}_3$, hence a 3-cycle, and $su \ne tu$, $us \ne ut$. So $X_3 u = u X_3 = c + c^2$, and $X_3(X_3 - u) = (X_3 - u)X_3 = 2$. □

*Proof of Theorem 3 (written).* Let $L$ be a lattice in $S^\lambda_{\mathbb{Q}}$ stable under the permutations of $\{1,2,3\}$. Both $X_3$ and $X_3 - u$ lie in $\mathbb{Z}\mathfrak{S}_3$, so in a $\mathbb{Z}$-basis of $L$ they act by integer matrices $A$ and $B$, and $AB = BA = 2I$ by Lemma 5. Choose unimodular $P, Q$ with $PAQ = D = \mathrm{diag}(d_1, \ldots, d_f)$, $f = f^\lambda$. Then $D \cdot (Q^{-1}BP^{-1}) = 2I$ with $Q^{-1}BP^{-1}$ an integer matrix, so $d_i C_{ii} = 2$ for an integer $C_{ii}$, and each $d_i$ is 1 or 2. The number of $d_i$ equal to 2 is the 2-adic valuation of $|\det A|$.

The determinant of $X_3$ on $S^\lambda_{\mathbb{Q}}$ does not depend on $L$. By the classical theorem of Jucys and Murphy [4, 6], $X_3$ acts diagonally on Young's seminormal basis of $S^\lambda_{\mathbb{Q}}$, with eigenvalue on the vector indexed by a standard tableau $T$ equal to the content of the cell of 3 in $T$. In a standard tableau the entries $1,2,3$ fill a partition of 3, so 3 occupies $(1,2)$, $(2,1)$, $(1,3)$ or $(3,1)$, of contents $1, -1, 2, -2$. Hence $|\det A| = 2^m$, and the Smith form is $(1^{f-m}, 2^m)$. □

*Proof of Corollary 4 (written).* For a cell $c$ let $N_c$ be the number of standard tableaux of shape $\lambda$ in which 3 occupies $c$ ($N_c = 0$ if $c \notin \lambda$), so that $m = N_{(1,3)} + N_{(3,1)}$. Every $S_r$ is contained in $\{(1,2),(2,1),(1,3),(3,1)\}$, and the first two cells have contents $\pm 1$. Hence the prediction is

$$\alpha_{f-r+1} = 2^{[N_{(1,3)} \ge r] + [N_{(3,1)} \ge r]} \qquad (1 \le r \le f),$$

where $[\,\cdot\,]$ is 1 if the condition holds and 0 otherwise.

Suppose $\lambda_1 \geq 3$ and $\lambda$ has at least three rows. The standard fillings of size three that place $1, 2, 3$ along the first row, or down the first column, put 3 in $(1, 3)$, respectively $(3, 1)$; both shapes lie in $\lambda$, and each filling extends to a standard tableau of shape $\lambda$ by filling the remaining cells with $4, \ldots, n$ row by row, top to bottom and left to right. So both contents 2 and $-2$ occur, $N_{(1,3)} \geq 1$ and $N_{(3,1)} \geq 1$, and the prediction gives $\alpha_f = 4$. By Theorem 3 every invariant factor divides 2, so the conjecture fails, on every lattice of the theorem.

Otherwise at most one of $(1, 3)$ and $(3, 1)$ lies in $\lambda$, and the corresponding $N_c$ equals $m$ (if neither does, $m = 0$). The prediction is then $\alpha_{f-r+1} = 2$ for $r \leq m$ and 1 for $r > m$, that is, $(1^{f-m}, 2^m)$, which is the Smith form given by Theorem 3. □

**Which steps are checked.** In the proof of Theorem 3, Lean checks the identity (`group_ring_identity`) and the matrix step: for integer matrices with $AB = 2I$, every unimodular diagonalization of $A$ has entries $\pm 1$ or $\pm 2$, and the number of even entries is $v_2(\det A)$ (`invariant_factors_of_mul_eq_two`). The rest is written: applying this to the nonnegative Smith form, that a lattice stable under the permutations of $\{1, 2, 3\}$ carries integer matrices $A$ and $B$ of $X_3$ and $X_3 - (1, 2)$ with $AB = 2I$ (an elementary consequence of the identity), the content spectrum of $X_3$ (classical, cited), and the description of the cells that 3 can occupy. In the proof of Corollary 4, Lean checks the final arithmetic (`k3_counting`): with $a = N_{(1,3)}$, $b = N_{(3,1)}$ and $a + b \leq f$, the list with $2^{[r \leq a] + [r \leq b]}$ in position $f - r + 1$ $(1 \leq r \leq f)$ equals $(1^{f-a-b}, 2^{a+b})$ if and only if $a = 0$ or $b = 0$. The facts about tableaux (which cells 3 can occupy, and the extension of the two fillings of size three) are written. The census of Section 7 agrees with both results for $n \leq 8$: in all 63 cases with $k = 3$ every invariant factor is 1 or 2, there are exactly $m$ factors equal to 2, and the disagreements are exactly the shapes with $\lambda_1 \geq 3$ and at least three rows.

*Remark* 6 (proved in writing; not machine-checked). Over $\mathbb{Z}_{(2)}$, the idempotent $P = \frac{1}{3}(1 + c + c^2)$ gives the same conclusion in another form: $X_3^2 = 1 + 3P$, so $X_3$ is invertible on $(1 - P)L_{(2)}$ and acts as $2(1, 3)$ on $PL_{(2)}$. Comparing the two descriptions gives $m = (f^\lambda + 2\chi^\lambda((1, 2, 3)))/3$, where $\chi^\lambda$ is the character of $S^\lambda_{\mathbb{Q}}$.

## 6 The cases with $n \leq 5$

**Proposition 7** (exact computation on the integral Specht lattices)**.** *On $S^\lambda_{\mathbb{Z}}$, Conjecture 7.4 holds for every $\lambda \vdash n \leq 4$ and every $1 \leq k \leq n$. For $n = 5$ it fails exactly for $\lambda = (3, 1, 1)$ with $k = 3$ and with $k = 5$.*

Supporting checks: this is a bounded exact computation by the standalone checker `check_small.py` in `checks/`, which recomputes all 69 cases with $n \leq 5$ from the definitions, proves the standard basis property in each, and computes Smith forms from determinantal divisors (Section 9). The census, a separate computation, agrees. For $\lambda = (3, 1, 1)$ and $k = 5$ the Smith form is $(1, 1, 2, 2, 4, 4)$ and the prediction $(1, 1, 1, 4, 4, 4)$. For $n \leq 4$ there is also a written argument, which gives more.

**Proposition 8** (proved in writing; not machine-checked)**.** *For every $\lambda \vdash n \leq 4$ and every $1 \leq k \leq n$, Conjecture 7.4 holds on every lattice in $S^\lambda_{\mathbb{Q}}$ preserved by $X_k$.*

Supporting checks: on the lattices $S^\lambda_{\mathbb{Z}}$ the statement is part of Proposition 7.

*Proof (written).* We use again that $X_k$ acts diagonally on Young's seminormal basis, with eigenvalues the contents of the cells occupied by $k$ in the standard tableaux [4, 6]. The determinant and the rational rank of $X_k - a$ ($a \in \mathbb{Z}$) therefore do not depend on the lattice.

If $k = 1$, then $X_1 = 0$, and 1 occupies $(1,1)$, of content 0, in every tableau; the prediction is $(0,\dots,0)$. If $k = 2$, then $X_2 = (1,2)$ is an involution, hence unimodular on every lattice, and its Smith form is $(1,\dots,1)$; the entry 2 occupies $(1,2)$ or $(2,1)$, of content $\pm 1$, so every predicted factor is 1. If $f^\lambda = 1$, then $\lambda = (n)$ or $(1^n)$, $X_k$ is the scalar $\pm(k-1)$, and $k$ occupies $(1,k)$ or $(k,1)$, as predicted. The remaining cases are listed below, with the cells occupied by $k$ in the standard tableaux (multiplicities as exponents) and the prediction.

| $\lambda$, $k$ | cells of $k$ | eigenvalues | prediction |
|---|---|---|---|
| $(2,1)$, 3 | $(1,2)$, $(2,1)$ | $1, -1$ | $(1,1)$ |
| $(2,2)$, 3 | $(1,2)$, $(2,1)$ | $1, -1$ | $(1,1)$ |
| $(2,2)$, 4 | $(2,2)^2$ | $0, 0$ | $(0,0)$ |
| $(3,1)$, 3 | $(1,3)$, $(1,2)$, $(2,1)$ | $2, 1, -1$ | $(1,1,2)$ |
| $(2,1,1)$, 3 | $(3,1)$, $(1,2)$, $(2,1)$ | $-2, 1, -1$ | $(1,1,2)$ |
| $(3,1)$, 4 | $(1,3)^2$, $(2,1)$ | $2, 2, -1$ | $(1,2,2)$ |
| $(2,1,1)$, 4 | $(3,1)^2$, $(1,2)$ | $-2, -2, 1$ | $(1,2,2)$ |

In the first two rows $|\det X_k| = 1$, so the Smith form is $(1,1)$. In the third, $X_4$ is diagonalizable with both eigenvalues 0, so $X_4 = 0$. In the fourth and fifth, $|\det X_3| = 2$ is squarefree, which forces $(1,1,2)$ (this also follows from Corollary 4). In the last two, $\lambda = (3,1)$ or $(2,1,1)$ with $k = 4$, we have $|\det X_4| = 4$. On a lattice preserved by $X_4$, the matrix of $X_4 - 2I$ for $(3,1)$, or of $X_4 + 2I$ for $(2,1,1)$, is an integer matrix of rational rank 1, since the eigenvalue $\pm 2$ has multiplicity 2 and $X_4$ is diagonalizable; all its $2 \times 2$ minors vanish, so it has rank at most 1 over $\mathbb{F}_2$. As $X_4 \equiv X_4 \mp 2I \pmod 2$, at least two invariant factors are even, and the product 4 forces $(1,2,2)$. Every case agrees with the prediction. □

## 7 The census for $n \le 8$

We computed the Smith form of $X_k$ on $S^\lambda_\mathbb{Z}$ for every $\lambda \vdash n \le 8$ and every $1 \le k \le n$, 416 cases in all, and compared it with Conjecture 7.4. The numbers of disagreements are:

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | total |
|---|---|---|---|---|---|---|---|---|---|
| cases | 1 | 4 | 9 | 20 | 35 | 66 | 105 | 176 | 416 |
| disagreements | 0 | 0 | 0 | 0 | 2 | 5 | 23 | 64 | 94 |

The two disagreements at $n = 5$ are those of Proposition 7. Stanley's example $\lambda = (5,1)$, $k = 5$, is reproduced: the Smith form is $(1,1,3,3,12)$, as the survey states. Of the 94 disagreements, 23 have $k = 3$ and are accounted for by Corollary 4. Disagreements involve the prime 2 in 79 cases and the prime 3 in 30; they occur both for $k < n$ (85 cases) and for $k = n$ (9 cases), and for two-row shapes from $n = 7$ on, the first being $\lambda = (5,2)$.

The weaker remark fails first at $n = 7$, for $(5,2)$ and $(2,2,1,1,1)$ with $k = 7$ (Proposition 2). Nor is the Smith form of $X_k$ always that of the diagonal matrix of its eigenvalues: this holds in 279 of the 416 cases.

What survives in the data is limited: the conjecture holds for all $n \le 4$ (Proposition 7) and at $k = 3$ for the shapes of Corollary 4, and the Smith forms of $\lambda$ and $\lambda'$ agree in every case. The modifications of the formula we tested all fail (Appendix C), and we propose no corrected form.

## 8 An open point

The documented facts are these. Stanley reports computations through $n \leq 7$, which he credits to Grinberg, and states that the conjecture was made on the basis of these data; the survey defines $W_{\lambda,k}$ through a basis of $\mathbb{Z}\mathfrak{S}_n$. Grinberg's later comment says that it is defined "somewhat clumsily" and names the integral Specht module with its standard basis as the intended lattice. On that lattice the conjecture fails at $n = 5$ (Theorem 1), the weaker remark fails at $n = 7$ (Proposition 2), and the census finds 30 disagreements with $n \leq 7$. The data of the computations reported in the survey are not available to us, and why they supported the conjecture is not known to us.

## 9 Verification

Three kinds of evidence are combined, and each result says which it has. Lean proves statements about explicit algebraic objects: the identity of Lemma 5; for the explicit matrices of Theorem 1 and Proposition 2, their Smith forms, the unreachability of the predicted forms, $AB = BA = 2I$ for the first and $A(A - 4I) = 0$ and the eigenvalues for the second; the matrix step of Theorem 3; and the counting step of Corollary 4. Standalone certificate checkers establish, by recomputation from the definitions, that these matrices are those of $X_k$ on the integral Specht lattices in the standard polytabloid basis, and that the predicted lists are Stanley's; together with Lean this verifies Theorem 1 and Proposition 2. A third checker verifies Proposition 7 by exact computation over the 69 cases with $n \leq 5$. Theorem 3, Corollary 4, Proposition 8, Remark 6, the derivation of Appendix B and the general statements about other lattices are proved in writing and are not machine-checked; the census is supporting data for them. The checks run with `checks/check.py` and `lake build` in `lean/`. Appendix D describes the checkers, the census construction and the Lean statements, with what each does not cover.

## A Quotations

The following is quoted from the final part of Section 7 of [8] (arXiv version 2, page 15; the text of version 1 is the same). Reference numbers inside quotations are those of the quoted source.

> One intriguing example is the *Jucys-Murphy element* (though it first appears in the work of Alfred Young [50, §19]) $X_k \in \mathbb{Q}\mathfrak{S}_n$, $1 \leq k \leq n$. It is defined by $X_1 = 0$ and
>
> $$X_k = (1,k) + (2,k) + \cdots + (k-1,k), \quad 2 \leq k \leq n,$$
>
> where $(i,k)$ denotes the transposition interchanging $i$ and $k$. Just as for $\Gamma_n(q)$, we can choose an integral basis for $\mathbb{Q}\mathfrak{S}_n$ (that is, a $\mathbb{Z}$-basis for $\mathbb{Z}\mathfrak{S}_n$) so that the action of $X_k$ on $\mathbb{Q}\mathfrak{S}_n$ with respect to this basis has a matrix of the form $\bigoplus_{\lambda \vdash n} f^\lambda W_{\lambda,k}$. The eigenvalues of $W_{\lambda,k}$ are known to be the contents of the positions occupied by $k$ in all standard Young tableaux of shape $\lambda$. [...] Darij Grinberg (private communication) computed the SNF of the matrices $W_{\lambda,k}$ for $\lambda \vdash n \leq 7$. On the basis of this data we make the following conjecture.
>
> CONJECTURE 7.4. *Let $\lambda \vdash n$, $1 \leq k \leq n$, and $W_{\lambda,k} \overset{\text{snf}}{\to} (\alpha_1, \ldots, \alpha_{f^\lambda})$. Fix $1 \leq r \leq f^\lambda$. Let $S_r$ be the set of positions $(i,j)$ that $k$ occupies in at least $r$ of the SYT's of shape $\lambda$. Then $\alpha_{f^\lambda - r + 1} = \pm \prod_{(i,j) \in S_r} (j - i)$.*
>
> Note in particular that every SNF diagonal entry is (conjecturally) a product of some of the eigenvalues of $W_{\lambda,k}$.

For example, when $\lambda = (5,1)$ and $k = 5$ we have $f^{(5,1)} = 5$ and $S_1 = \{(1,5),(2,1),(1,4)\}$, $S_2 = S_3 = \{(1,4)\}$, $S_4 = S_5 = \emptyset$. Hence $W_{(5,1),5} \overset{\text{snf}}{\to} (1,1,3,3,12)$.

The omitted passage lists the standard Young tableaux of shape $(5,1)$ and their eigenvalues.

In his comment of December 25, 2024 on [1], Grinberg wrote (the emphasis, bold in the original, is set in italics here):

> Note about Question 1: The $W_{\lambda,k}$ in §7 of Stanley's paper is defined somewhat clumsily. It should be defined as the matrix representing the action of the YJM element $X_k$ on the Specht module $\mathcal{S}^\lambda$ over $\mathbb{Z}$ with respect to the standard tableau basis. You can, of course, take any other basis, since the Smith normal form is conjugation-invariant. But the direct sum $\bigoplus_{\lambda \vdash n} f^\lambda W^{\lambda,k}$ is *not* the action of $X_k$ on the group ring of $S_n$, so you cannot define it through this sum.

Question 1 of [1], to which the note refers, reads: "Is there any progress on these conjectures ?" In a second comment Grinberg added that he was "still interested in Question 1."

## B A derivation by hand for $\lambda = (3,1,1)$

This derivation is written and not machine-checked. The matrix can also be obtained without coordinates in the tabloid module. Let $e_1, \dots, e_5$ be the standard basis of $\mathbb{Z}^5$, with $\mathfrak{S}_5$ permuting it, and let $A_4 = \{x \in \mathbb{Z}^5 : \sum_i x_i = 0\}$. The map $\varphi$ sending the tabloid with $x$ in row 2 and $y$ in row 3 to $e_x \wedge e_y \in \Lambda^2 \mathbb{Z}^5$ is $\mathfrak{S}_5$-equivariant. If $T$ has first column $(c_1, c_2, c_3)$, then $C_T$ is the symmetric group on $\{c_1, c_2, c_3\}$, and summing the six signed terms gives

$$\varphi(e_T) = 2\,(e_{c_2} - e_{c_1}) \wedge (e_{c_3} - e_{c_1}) \in 2\Lambda^2 A_4.$$

For standard $T$ the first column is $(1, a, b)$ with $2 \le a < b \le 5$, each pair occurring once, so the six standard polytabloids are sent to $2 f_a \wedge f_b$, where $f_i = e_i - e_1$. These form twice a $\mathbb{Z}$-basis of $\Lambda^2 A_4$. Since $S^\lambda_{\mathbb{Q}}$ is irreducible and $\varphi \neq 0$, the map is injective on it, and $\frac{1}{2}\varphi$ is an $\mathfrak{S}_5$-equivariant isomorphism $S^{(3,1,1)}_{\mathbb{Z}} \cong \Lambda^2 A_4$. Now use the basis $b_i = e_i - e_5$ $(1 \le i \le 4)$ of $A_4$ and write $b_{ij} = b_i \wedge b_j$. The transpositions $(1,3)$ and $(2,3)$ fix $b_4$ and permute $b_1, b_2, b_3$, so $X_3$ preserves the span of $b_{12}, b_{13}, b_{23}$ and the span of $b_{14}, b_{24}, b_{34}$, with matrices

$$\begin{pmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & -1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Their determinants are 2 and $-2$, and each has a $2 \times 2$ minor equal to $\pm 1$, so each has Smith form $(1,1,2)$. Together they give $(1,1,1,1,2,2)$ again.

## C Other readings and modifications of the formula

**Other lattices for the counterexample (written; not machine-checked).** By Theorem 3, the same Smith form arises on any lattice stable under the permutations of $\{1,2,3\}$ in a module isomorphic to $S^{(3,1,1)}_{\mathbb{Q}}$: for instance the lattices given by polytabloids built from column tabloids, by a left ideal generated by a Young symmetrizer, by the dual lattice, or by the saturated lattice $S^\lambda_{\mathbb{Q}} \cap M^\lambda_{\mathbb{Z}}$. No choice of lattice or basis of this kind restores the prediction.

**Other lattices in the census.** For $k = 3$, Theorem 3 (proved in writing) covers every lattice stable under the permutations of $\{1, 2, 3\}$. Over the whole census the following were also compared. The dual lattice $\mathrm{Hom}(S^\lambda_\mathbb{Z}, \mathbb{Z})$, with the contragredient action, carries the transposed matrix, since transpositions are their own inverses; its Smith form is the same. The saturated lattice $S^\lambda_\mathbb{Q} \cap M^\lambda_\mathbb{Z}$, which contains $S^\lambda_\mathbb{Z}$, gives the same Smith form as $S^\lambda_\mathbb{Z}$ in all 135 cases with $n \leq 6$; it was not computed for larger $n$. The census also finds that $\lambda$ and its conjugate $\lambda'$ have the same Smith form for every $k$, in all 416 cases, as one expects from the classical relation between $S^{\lambda'}$ and the dual of $S^\lambda$ twisted by the sign character [3, §8], under which $X_k$ becomes $-X_k$. The statements about the dual lattice and this relation are written background, not machine-checked; the comparisons are finite computations.

**The literal group ring reading.** Read literally, the survey compares the Smith form of left multiplication by $X_k$ on $\mathbb{Z}\mathfrak{S}_n$ with that of a block diagonal matrix holding $f^\lambda$ copies of each predicted list. Grinberg's comment explains why this is not the intended statement, and it fails already for $n = 4$, $k = 4$. Writing $d^m$ for $m$ invariant factors equal to $d$, left multiplication by $X_4$ on $\mathbb{Z}\mathfrak{S}_4$ has invariant factors forming the multiset $\{0^4, 1^{16}, 2^4\}$. The predicted lists are (3) for (4) and $(1^4)$, $(1, 2, 2)$ for $(3, 1)$ and $(2, 1, 1)$, and $(0, 0)$ for $(2, 2)$; with multiplicities $f^\lambda$ the diagonal holds the multiset $\{0^4, 1^6, 2^{12}, 3^2\}$, whose Smith normalization has invariant factors forming the multiset $\{0^4, 1^8, 2^{10}, 6^2\}$. For $n \leq 3$ the two agree.

**Modifications of the formula.** We tested modifications of the formula against the census, the Smith forms of $X_k$ on $S^\lambda_\mathbb{Z}$ for all $\lambda \vdash n \leq 8$. Write $N_c$ for the number of standard tableaux of shape $\lambda$ in which $k$ occupies the cell $c$, so that the formula as stated uses $S_r = \{c : N_c \geq r\}$. All sets of cells below consist of cells that $k$ occupies, that is, cells with $N_c \geq 1$. The readings are:

- *as stated*: $\alpha_{f-r+1} = |\prod_{c \in S_r} \mathrm{content}(c)|$;
- *multiset*: the Smith form of the diagonal matrix of these $f$ products, ignoring their order;
- $\alpha_r$ *indexing*: the $r$-th product assigned to $\alpha_r$ instead of $\alpha_{f-r+1}$;
- *exactly $r$, more than $r$, at most $r$*: $S_r$ replaced by $\{c : N_c = r\}$, $\{c : N_c > r\}$ or $\{c : 1 \leq N_c \leq r\}$;
- *multiplicity classes*: one product for each positive value of $N_c$, over the cells with that value, the list padded with 1s to length $f$ and assigned in increasing or decreasing order of $N_c$, or Smith-normalized;
- *eigenvalue diagonal*: the Smith form of the diagonal matrix of all eigenvalues of $X_k$ on $S^\lambda_\mathbb{Q}$;
- *zero contents dropped*: the formula as stated, with cells of content 0 left out of the products, except that a nonempty set consisting only of cells of content 0 gives 0 (the empty set gives 1);
- $X_{k\pm1}$: the formula for $k$ compared with the Smith form of $X_{k+1}$ or of $X_{k-1}$;
- $Y_k$: the formula for $k$ compared with the Smith form of $Y_k = \sum_{i>k}(i, k)$;
- $\lambda'$: the formula for the conjugate shape $\lambda'$ compared with $X_k$ on $S^\lambda$;
- *fillings of* $1, \ldots, k$: $N_c$ replaced by the number of distinct restrictions to the entries $1, \ldots, k$ of standard tableaux of shape $\lambda$ with $k$ in $c$;
- *weak form*: every nonzero invariant factor is a product of some of the absolute values of the eigenvalues, the remark following the conjecture.

A case matches when the predicted list equals the Smith form entry by entry. The actual Smith form is written nonnegative, its nonzero entries in increasing order and its zeros last. For the formula as stated, the $\lambda'$, $X_{k\pm1}$ and $Y_k$ comparisons and the fillings reading, the predicted list is put in the same order before the comparison. The ordered trial lists of the other readings ($\alpha_r$ indexing, exactly, more than or at most $r$, multiplicity classes, zero contents dropped) keep their assigned positions $\alpha_1, \ldots, \alpha_f$, zeros included; their Smith-normalized versions are the Smith forms of the corresponding diagonal matrices. Where a reading has an ordered and a Smith-normalized version, the table gives the better count. The columns give the matches among the 240 cases with $n \le 7$ and the 416 with $n \le 8$ (196 and 350 for $X_{k\pm1}$, where $k \pm 1$ must lie in $1, \ldots, n$), whether the reading reproduces Stanley's example $(1, 1, 3, 3, 12)$ for $\lambda = (5, 1)$, $k = 5$, and the first case, in order of $n$, where it fails.

| reading | $n \le 7$ | $n \le 8$ | $(5,1)$, $k=5$ | first failure |
|---|---|---|---|---|
| as stated | 210 | 322 | yes | $(3,1,1),\ k=3$ |
| multiset | 210 | 322 | yes | $(3,1,1),\ k=3$ |
| $\lambda'$ | 210 | 322 | yes | $(3,1,1),\ k=3$ |
| $\alpha_r$ indexing | 122 | 181 | no | $(3,1),\ k=3$ |
| exactly $r$ | 96 | 133 | no | $(2,1),\ k=1$ |
| more than $r$ | 47 | 70 | no | $(1),\ k=1$ |
| at most $r$ | 94 | 131 | no | $(2,1),\ k=1$ |
| multiplicity classes | 96 | 133 | no | $(2,1),\ k=1$ |
| eigenvalue diagonal | 177 | 279 | yes | $(3,2),\ k=4$ |
| zero contents dropped | 173 | 266 | yes | $(3,2),\ k=4$ |
| $X_{k+1}$ | 3 | 3 | no | $(2),\ k=1$ |
| $X_{k-1}$ | 3 | 3 | no | $(2),\ k=2$ |
| $Y_k$ | 26 | 26 | no | $(2),\ k=1$ |
| fillings of $1, \ldots, k$ | 144 | 203 | yes | $(2,1),\ k=1$ |
| weak form | 238 | 410 | yes | $(5,2),\ k=7$ |

The multiset reading agrees with the formula as stated in every case, since $S_1 \supseteq S_2 \supseteq \cdots$ makes the stated products a divisibility chain. The $\lambda'$ reading agrees with it as well: conjugation moves the counts to transposed cells and changes the sign of the contents, and the census finds the same Smith form for $\lambda$ and $\lambda'$. So does the reading with the labels counted from $n$ downward, $Y_{n+1-k} = w_0 X_k w_0$ with $w_0$ the reversal permutation, whose Smith form equals that of $X_k$ in all 416 cases.

The modifications we tested all fail on the data with $n \le 7$. The best fit is the formula as stated, together with its equivalent readings; it matches 210 of the 240 cases with $n \le 7$ and first fails at $\lambda = (3, 1, 1)$, $k = 3$. Every other reading that predicts the whole Smith form first fails at $n \le 5$, most of them at $n \le 3$. The weak form, which predicts only a property of it, holds for all $n \le 6$ and fails at $n = 7$, for $(5, 2)$ and $(2, 2, 1, 1, 1)$ with $k = 7$ (Proposition 2). We propose no corrected form. The program for this comparison is included with the optional census programs; it needs the python-flint library.

## D Scope of the checks

**Certificates and checkers.** The two counterexamples, $\lambda = (3, 1, 1)$ with $k = 3$ and $\lambda = (5, 2)$ with $k = 7$, are given as certificates, each holding the standard tableaux in order, the matrix $A$ of $X_k$, unimodular $U$ and $V$ with $UAV$ diagonal, the Smith form and the prediction. Two standalone checkers, written independently and using only the Python standard library, read them; neither

trusts the encoded matrix. The first enumerates the standard tableaux from all $n!$ fillings, proves that the standard polytabloids form a $\mathbb{Z}$-basis of the $\mathbb{Z}$-span of all $n!$ polytabloids, recomputes $A$ exactly from the tabloid module, checks $UAV = \mathrm{diag}(\mathrm{snf})$ with integer $U, V$ of determinant $\pm 1$ and a divisibility chain, and recomputes the prediction. The second, with separate code, enumerates the standard tableaux by lattice words, shows that the $\mathbb{F}_2$-span of all polytabloids has dimension $f^\lambda$, computes the rank of $X_k$ there without reference to a basis, compares it with the number of odd predicted factors, verifies $X_k e_{T_j} = \sum_i A_{ij} e_{T_i}$ in the tabloid module for every $j$, and checks $U$, $V$ and the Smith form with its own arithmetic. For a certificate that also claims the failure of the weak remark, both checkers, each with its own code, take as eigenvalues the contents of the cell holding $k$ in each standard tableau; verify $\prod_c (A - cI) = 0$ over the distinct contents $c$, so that $A$ is diagonalizable with eigenvalues among them; check $\mathrm{tr}(A^j) = \sum c^j$ over the eigenvalue list for $1 \le j$ below the number of distinct contents, which fixes the multiplicities; and require some Smith entry that is not the absolute value of a product of a sub-multiset of the eigenvalues (the empty product being 1). The checkers are independent: separate code, different enumerations of the standard tableaux (all $n!$ fillings, and lattice words), and different arithmetic (exact rational elimination, and Bareiss determinants with linear algebra over $\mathbb{F}_2$). Both reject eight corrupted certificates (a changed entry of $A$; the Smith form replaced by the prediction; a non-unimodular $U$; a rational $U$ of determinant $\pm 1$; an extra factor; a non-standard basis tableau; an altered prediction; and, in the weak-form certificate, a planted change to $A$ with $U$ adjusted so that $UAV$ is still the claimed diagonal matrix, rejected because $A(A-4I) \neq 0$). Both also reject three controls: the agreeing case $\lambda = (5,1)$, $k = 5$, as showing no discrepancy, and two weak-form claims about true statements, $(5,1)$ with $k = 5$, where $12 = 4 \cdot 3$ is a product of eigenvalues, and $(3,1,1)$ with $k = 3$, where $2 = 2 \cdot 1$ is one. A third checker, written separately from both and importing neither, recomputes every $\lambda \vdash n \le 5$ and every $k$: it proves that the standard polytabloids form a $\mathbb{Z}$-basis, forms the matrix of $X_k$, computes its Smith form from determinantal divisors, and asserts Proposition 7 and Stanley's example $\lambda = (5,1)$, $k = 5$. The entry point is `checks/check.py`, with the certificates in `data/`.

**Census.** The census was computed by two constructions. The first generates $S^\lambda_\mathbb{Z}$ inside $M^\lambda_\mathbb{Z}$ as the $\mathbb{Z}\mathfrak{S}_n$-span of one polytabloid, closing under simple transpositions with Hermite normal forms; it uses neither the standard basis nor straightening, and every Smith form is computed by two implementations. The second uses the standard polytabloid basis and exact rational linear algebra. They agree in all 135 cases with $n \le 6$. The census output is included as `data/census-n8.json`; the census programs are optional and not part of the check.

**Lean.** The checked statements are in Lean 4 [2] with mathlib [5], in the namespace `StanleySmithForms`.

- `group_ring_identity`: the identity of Lemma 5 in the integral group ring of $\mathfrak{S}_n$, for every $n \ge 3$.
- `A_det`: $\det A = -4$; `U_det`, `V_det`: $\det U = -1$, $\det V = 1$; `smith_form`: $UAV = \mathrm{diag}(1,1,1,1,2,2)$.
- `A_mul_B`, `B_mul_A`: $AB = BA = 2I$ for the explicit integer matrix $B = 2A^{-1}$.
- `rank_mod_two`: $A$ has rank 4 over $\mathbb{F}_2$; `not_predicted`: no integer matrices $P, Q$ with unit determinant satisfy $PAQ = \mathrm{diag}(1,1,1,1,1,4)$; `counterexample` combines the Smith form and the unreachability of the predicted form.

- `invariant_factors_of_mul_eq_two`: for integer matrices $A, B$ with $AB = 2I$, every unimodular diagonalization $UAV = \mathrm{diag}(d)$ has $d_i \in \{\pm 1, \pm 2\}$, and the number of even $d_i$ is $v_2(\det A)$ (applying it to the nonnegative Smith form of an arbitrary lattice is the written step of Theorem 3); `witness_factors` applies it to the witness.
- For $\lambda = (5, 2)$, $k = 7$, with the explicit $14 \times 14$ matrix $A$: `smith_form_52`, `U52_unimodular`, `V52_unimodular`: the Smith form via unimodular $U$, $V$; `eigen_identity_52`: $A(A - 4I) = 0$; `charpoly_52`: the characteristic polynomial of $A$ is $x^5(x-4)^9$; `eigenvalues_52`: the rational eigenvalues of $A$ are exactly 0 and 4; `weak_remark_fails_52`: every unimodular diagonalization of $A$ has an entry divisible by 2 but not by 4, proved from a parity certificate, integer vectors $x, y, w, z$ with $Ay = 2x$, $w^{\mathsf{T}}A = 2z^{\mathsf{T}}$ and $w \cdot x = 1$; `counterexample_52` combines the Smith form, the eigenvalues and this statement. The identities between explicit $14 \times 14$ matrices are checked by evaluation in the Lean kernel, which adds no axioms.
- `k3_counting`: the counting step of Corollary 4, as stated in Section 5.

Lean does not construct Specht modules, Young's seminormal basis or standard tableaux. It does not check that the explicit matrices are those of $X_3$ on $S^{(3,1,1)}_{\mathbb{Z}}$ and of $X_7$ on $S^{(5,2)}_{\mathbb{Z}}$, nor that the predicted lists and the contents are Stanley's; the certificate checkers, which recompute these from $\lambda$ and $k$, cover that step for both counterexamples. It does not check the passage from the group ring to an arbitrary lattice, the content spectrum, or the facts about tableaux used in Theorem 3 and Corollary 4; those are written.

## References


[1] A. Chervov, *Conjectures on Smith Normal Form for Jucys-Murphy elements (Stanley, Grinberg) - status and generalisations?*, with comments by D. Grinberg, MathOverflow question 484782, December 25, 2024, https://mathoverflow.net/questions/484782.

[2] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction – CADE 28*, Lecture Notes in Comput. Sci. 12699, Springer, 2021, 625–635.

[3] G. D. James, *The Representation Theory of the Symmetric Groups*, Lecture Notes in Math. 682, Springer, Berlin, 1978.

[4] A.-A. A. Jucys, Symmetric polynomials and the center of the symmetric group ring, *Rep. Math. Phys.* **5** (1974), 107–112.

[5] The mathlib Community, The Lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, ACM, 2020, 367–381.

[6] G. E. Murphy, A new construction of Young's seminormal representation of the symmetric groups, *J. Algebra* **69** (1981), 287–297.

[7] B. E. Sagan, *The Symmetric Group: Representations, Combinatorial Algorithms, and Symmetric Functions*, 2nd ed., Grad. Texts in Math. 203, Springer, New York, 2001.

[8] R. P. Stanley, Smith normal form in combinatorics, *J. Combin. Theory Ser. A* **144** (2016), 476–495; arXiv:1602.00166.